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Studierichting: Informatica

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$$f_n : [1, +\infty[\rightarrow \mathbb{R} : x \rightarrow \frac{x^n}{x^n + 1}$$

$$\lim_{n \rightarrow \infty} \frac{x^n}{x^n + 1} = \begin{cases} 1 & \text{als } x > 1 \\ \frac{1}{2} & \text{als } x = 1 \end{cases}$$

$$f : [1, +\infty[\rightarrow \mathbb{R} : x \rightarrow \begin{cases} 1 & \text{als } x > 1 \\ \frac{1}{2} & \text{als } x = 1 \end{cases} \quad \text{en } f_n \xrightarrow{p} f$$

$\Rightarrow f_n \not\xrightarrow{u} f$ want f is discontinu

References